GEOMETRIC NORMALITY AT A POINT IS UNCHANGED BY AN ARBITRARY EXTENSION OF THE BASE FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the pointwise criterion; no step is referred to the reader. The author's parenthetical that the point over x is unique for a purely inseparable extension is reproduced verbatim.

Editorial note. The counted spans carry 3 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-geometrically-normal-upstairs is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Normality descends along faithfully flat local homomorphisms. Algebra, Lemma algebra-lemma-descent-normal, invoked at proof lines 2237-2238. Supplies that normality of the local ring upstairs forces normality downstairs.
- Existence of a common field extension of two given extensions. Fields, Lemma fields-lemma-common-extension-field, invoked at proof lines 2228-2229. Supplies the field k"' and the point lying above both x' and x".
- Pointwise criterion for geometric normality. Same chapter, Lemma lemma-geometrically-normal-at-point, invoked at proof lines 2239-2240. Supplies the passage from finite purely inseparable base changes to geometric normality over k.

Source reference. Original lemma 033G: sources/stacks-chow/tags.txt, source lines 4055–4055. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0H7K: sources/stacks-chow/tags.txt, source lines 21460–21460. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 038N: sources/stacks-varieties/pinned-original-source.tex, source lines 2118–2118. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let k be a field. Let X be a scheme over k. Let k'/k be a field extension. Let $x \in X$ be a point, and let $x' \in X_{k'}$ be a point lying over x. The following are equivalent*

(1) *X is geometrically normal at x,*
(2) *$X_{k'}$ is geometrically normal at x'.*

In particular, X is geometrically normal over k if and only if $X_{k'}$ is geometrically normal over k'.

Proof. It is clear that (1) implies (2). Assume (2). Let k''/k be a finite purely inseparable field extension and let $x'' \in X_{k''}$ be a point lying over x (actually it is unique). Choose a common field extension k'''/k of k'/k and k''/k, see Fields, Lemma 0H7K. Choose a point $x''' \in X_{k'''}$ lying over both x' and x''. Consider the map of local rings

$$\mathcal{O}_{X_{k''},x''} \longrightarrow \mathcal{O}_{X_{k'''},x'''}.$$

This is a flat local ring homomorphism and hence faithfully flat. By (2) we see that the local ring on the right is normal. Thus by Algebra, Lemma 033G we conclude that $\mathcal{O}_{X_{k''},x''}$ is normal. By Lemma 038N we see that X is geometrically normal at x. □